Local matrix-mass bounds do not imply Stahl–Totik regularity

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Abstract

We construct a normalized nontrivial positive-semidefinite matrix measure with a uniform polynomial local-mass lower bound whose block-Jacobi determinant regularity limit nevertheless vanishes. The rank-one density has no polynomial null vector. This gives a counterexample to the precise matrix local-mass criterion in AIM 1035.

Repository target. This manuscript concerns AIM problem 1035 at revision `3fa50884d129`. Its scope is stated below.

Status and scope

This counterexample disproves AIM 1035 and Conjecture 5.9 in the work of D. Damanik, A. Pushnitski and B. Simon, *The Analytic Theory of Matrix Orthogonal Polynomials*, Surveys in Approximation Theory **4** (2008), 1–85 [1]. It uses the positive-semidefinite matrix measures allowed by that paper. The polynomial moment form is strictly positive, as required by [1, Section 1.2 and Lemma 2.1]. The density has rank one, so a variant requiring a positive-definite density almost everywhere is not addressed.

Primary published text: https://www.math.auckland.ac.nz/sat/papers/11/11.pdf. The relevant source equations are (2.34) and (5.20); the needed facts are proved below.

Theorem

There exists a normalized, nontrivial, positive 2×2 matrix measure μ, supported on $E = [-1, 1]$, whose associated block Jacobi matrix has essential spectrum exactly E, such that

$$\mu([x - m^{-1}, x + m^{-1}]) \succeq c m^{-3} I_2 \qquad (x \in E,\ m \geq 1)$$

for a constant $c > 0$, but whose invertible off-diagonal Jacobi blocks satisfy

$$\lim_{n \to \infty} |\det(A_1 \cdots A_n)|^{1/n} = 0.$$

Since $\mathrm{Cap}([-1, 1])^2 = 1/4$, this measure is not regular. Its mass condition is stronger than the conjectured condition: for every $\eta > 0$, one has $E \subset S_{m,\eta}$ for all sufficiently large m.

1. Construction and nondegeneracy

Set

$$v(x) = \begin{pmatrix} 1 \\ e^x \end{pmatrix}, \qquad d\nu(x) = v(x)v(x)^T \, \mathbf{1}_{[-1,1]}(x)\, dx.$$

Let $M = \nu(E)$, $C = M^{-1/2}$, and $d\mu = C\, d\nu\, C$. In particular, $\mu(E) = I_2$. Explicitly,

$$M = \begin{pmatrix} 2 & e - e^{-1} \\ e - e^{-1} & (e^2 - e^{-2})/2 \end{pmatrix}, \qquad \det M = 2(1 - e^{-2}) > 0.$$

For a vector polynomial $f = (a, b)^T$,

$$\|f\|_\nu^2 = \int_{-1}^{1} |a(x) + e^x b(x)|^2 \, dx.$$

This norm is zero only if $a = b = 0$. Indeed, vanishing of the integral forces the entire function $a(z) + e^z b(z)$ to vanish identically. If $b \neq 0$, its exponential term dominates the polynomial a on the positive real axis, a contradiction. The invertible constant change $f \mapsto Cf$ gives the same nondegeneracy for μ.

Consequently every finite vector-polynomial Gram matrix is positive definite. Block Gram–Schmidt therefore produces the monic matrix orthogonal polynomials of all degrees and an associated block Jacobi matrix with all off-diagonal blocks invertible. The rank-one density is compatible with strict positivity of the polynomial moment form: there is no nonzero polynomial vector lying in its pointwise nullspace.

2. The associated essential spectrum

Define, initially on vector polynomials,

$$(Uf)(x) = v(x)^T C f(x).$$

By construction, U is an isometry from the polynomial-vector inner product for μ into $L^2([-1,1], dx)$. Its range contains every scalar polynomial: for a scalar polynomial r, take $f = C^{-1}(r, 0)^T$. Scalar polynomials are dense in this scalar L^2 space. Thus U extends to a unitary map from the completion of the polynomial vectors onto $L^2([-1,1], dx)$.

Moreover, $U(xf) = xUf$. Hence the block Jacobi operator obtained from the orthonormal polynomial-vector basis is unitarily equivalent to multiplication by x on scalar $L^2([-1,1], dx)$. This is a bounded self-adjoint operator of norm one, with spectrum and essential spectrum both $[-1, 1]$. (Its spectrum is purely absolutely continuous of multiplicity one; the conjecture does not assume maximal matrix spectral multiplicity.)

3. Uniform local matrix-mass lower bound

Let $I \subset [-1, 1]$ be an interval of length $L > 0$. Write

$$G_I = \nu(I) = \begin{pmatrix} L & \int_I e^t \, dt \\ \int_I e^t \, dt & \int_I e^{2t} \, dt \end{pmatrix}.$$

The determinant identity

$$\det G_I = \frac{1}{2}\int_I\int_I (e^s - e^t)^2\,ds\,dt$$

and the mean-value bound $|e^s - e^t| \geq e^{-1}|s-t|$ imply

$$\det G_I \geq \frac{e^{-2}}{2}\int_I\int_I (s-t)^2\,ds\,dt = \frac{e^{-2}}{12}L^4.$$

Also $\operatorname{tr} G_I \leq (1+e^2)L$. Since G_I is positive definite,

$$\lambda_{\min}(G_I) = \frac{\det G_I}{\lambda_{\max}(G_I)} \geq \frac{e^{-2}}{12(1+e^2)}L^3.$$

For $x \in [-1,1]$ and $m \geq 1$, the truncated interval $I = [x-m^{-1}, x+m^{-1}] \cap [-1,1]$ has length at least m^{-1}, including when x is an endpoint. Consequently,

$$\mu([x-m^{-1}, x+m^{-1}]) = CG_IC \succeq \frac{e^{-2}}{12(1+e^2)\lambda_{\max}(M)}m^{-3}I_2.$$

Take the positive constant on the right as c. Because $cm^{-3} \geq e^{-\eta m}$ for every sufficiently large m, for each fixed $\eta > 0$, the required sets contain all of E eventually.

4. An elementary Padé remainder identity

For every integer $n \geq 1$, there are real polynomials q_n, p_n such that q_n is monic of degree n, $\deg p_n \leq n-1$, and

$$e^x q_n(x) - p_n(x) = \frac{x^{2n}}{(n-1)!}\int_0^1 e^{tx}t^n(1-t)^{n-1}\,dt. \tag{1}$$

Here is a direct proof. Let $f_n(t) = t^n(1-t)^{n-1}$. Integrating by parts repeatedly, using $f_n^{(2n)} = 0$, gives

$$\frac{x^{2n}}{(n-1)!}\int_0^1 e^{tx}f_n(t)\,dt = \frac{1}{(n-1)!}\sum_{j=0}^{2n-1}(-1)^j x^{2n-1-j}\big(e^x f_n^{(j)}(1) - f_n^{(j)}(0)\big).$$

The endpoint vanishing orders are $n-1$ at 1 and n at 0. Therefore the coefficient of e^x is a polynomial of degree n, while the other polynomial has degree at most $n-1$. Since $f_n^{(n-1)}(1) = (-1)^{n-1}(n-1)!$, the former polynomial is monic. This proves (1), initially for $x \neq 0$ and then at 0 by continuity.

For reference, explicit coefficients are

$$q_n(x) = \sum_{k=0}^{n}(-1)^{n+k}\frac{(2n-1-k)!\,n!}{(n-1)!\,(n-k)!\,k!}x^k,$$

$$p_n(x) = (-1)^n\sum_{k=0}^{n-1}\frac{(2n-1-k)!}{(n-1-k)!\,k!}x^k.$$

These are the monically rescaled Padé approximants of type $[n-1/n]$ for the exponential. No Padé approximation theorem is needed: identity (1) is sufficient.

For $|x| \le 1$, the beta integral yields

$$|e^x q_n(x) - p_n(x)| \le \frac{e}{(n-1)!} \int_0^1 t^n (1-t)^{n-1}\, dt = e\frac{n!}{(2n)!}. \tag{2}$$

5. The determinant regularity limit is zero

Let P_n denote the monic matrix orthogonal polynomial for μ, and let

$$H_n = \int P_n(x)^*\, d\mu(x)\, P_n(x).$$

For every monic matrix polynomial Q of degree n, orthogonality gives

$$\int Q^*\, d\mu\, Q = H_n + \int (Q - P_n)^*\, d\mu\, (Q - P_n) \succeq H_n. \tag{3}$$

Consider the monic trial matrix

$$T_n(x) = \begin{pmatrix} x^n & -p_n(x) \\ 0 & q_n(x) \end{pmatrix}$$

for ν, and $\widetilde{T}_n = C^{-1} T_n C$ for μ. Both have leading coefficient I_2. If $R_n = e^x q_n - p_n$, then

$$K_n := \int T_n^*\, d\nu\, T_n = \int_{-1}^1 \begin{pmatrix} x^{2n} & x^n R_n(x) \\ x^n R_n(x) & R_n(x)^2 \end{pmatrix} dx.$$

The determinant of this Gram matrix is bounded above by the product of its diagonal entries. By (2),

$$\det K_n \le \left(\int_{-1}^1 x^{2n}\, dx \right) \left(\int_{-1}^1 R_n(x)^2\, dx \right) \le 4e^2 \left(\frac{n!}{(2n)!} \right)^2. \tag{4}$$

The normalized trial norm is CK_nC. Since determinants are monotone for the positive-definite order, (3)–(4) imply

$$\det H_n \le \det(CK_nC) \le \frac{4e^2}{\det M} \left(\frac{n!}{(2n)!} \right)^2. \tag{5}$$

For a normalized matrix measure, comparison of the leading coefficients in the orthonormal three-term recurrence gives

$$\det H_n = |\det(A_1 \cdots A_n)|^2. \tag{6}$$

For completeness, the recurrence gives

$$|\det A_j| = (\det H_j / \det H_{j-1})^{1/2}.$$

Multiplying and using $H_0 = I_2$ gives (6). This is also [1, Eq. (2.34)].

Combining (5) and (6),

$$0 < |\det(A_1 \cdots A_n)| \le \frac{2e}{\sqrt{\det M}} \frac{n!}{(2n)!}.$$

Finally,

$$\frac{n!}{(2n)!} = \frac{1}{(n+1)(n+2)\cdots(2n)} \leq (n+1)^{-n}.$$

Taking n-th roots proves that the determinant limit is zero. This contradicts regularity, whose required value is $1/4$, and completes the counterexample.

Why the scalar intuition fails

Every fixed vector direction receives substantial mass on every short interval. Nevertheless, the null direction of the rank-one density rotates analytically with x. Vector polynomials can follow this changing null direction to superexponential accuracy, via rational approximation of e^x. The local integral lower bound does not control such changing polynomial directions. This mechanism is absent from a scalar measure or a fixed diagonal direct sum.

References

[1] D. Damanik, A. Pushnitski and B. Simon, *The Analytic Theory of Matrix Orthogonal Polynomials*, Surveys in Approximation Theory **4** (2008), 1–85.
https://www.math.auckland.ac.nz/sat/papers/11/11.pdf.
https://arxiv.org/abs/0711.2703v2. The positive-semidefinite convention is in Section 1.2 (p. 5), polynomial nontriviality in Lemma 2.1 (p. 16), the norm identity in (2.34) (p. 21), determinant regularity in Theorem 5.4 (p. 70), and Conjecture 5.9 in (5.20) (p. 72).